Singular Value Decomposition
SVD

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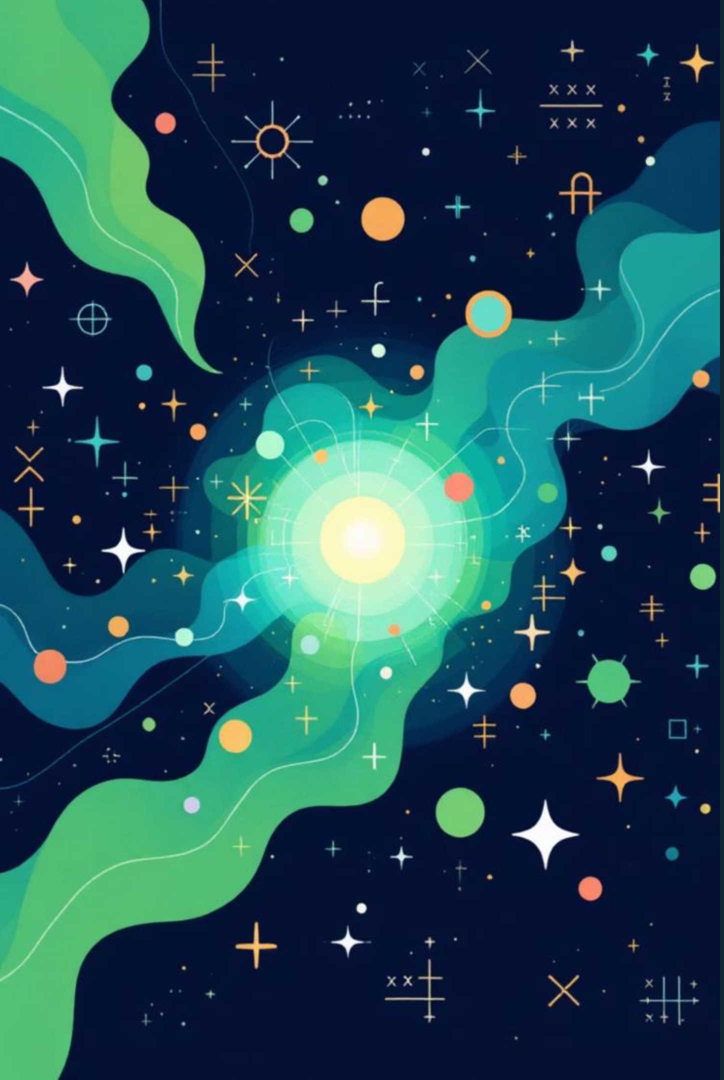


Singular Value Decomposition (SVD): Unlocking the Power of Matrix Factorization

Singular Value Decomposition (SVD) is a powerful mathematical technique with widespread applications across various scientific and engineering disciplines. It serves as a fundamental tool for dissecting and understanding complex datasets by breaking down matrices into simpler, interpretable components. This presentation explores the core concepts of SVD, its underlying mechanics, and its profound impact on real-world problems, from data compression to recommender systems.

What is SVD? A Geometric Intuition

At its heart, any matrix can be viewed as a linear transformation. This transformation isn't just a jumble of numbers; it's a dynamic from one space and maps them to another. Geometrically, SVD reveals that this complex transformation can always be broken down fundamental, sequential operations:

1

Rotation

An initial rotation of the input vectors.

2

Scaling

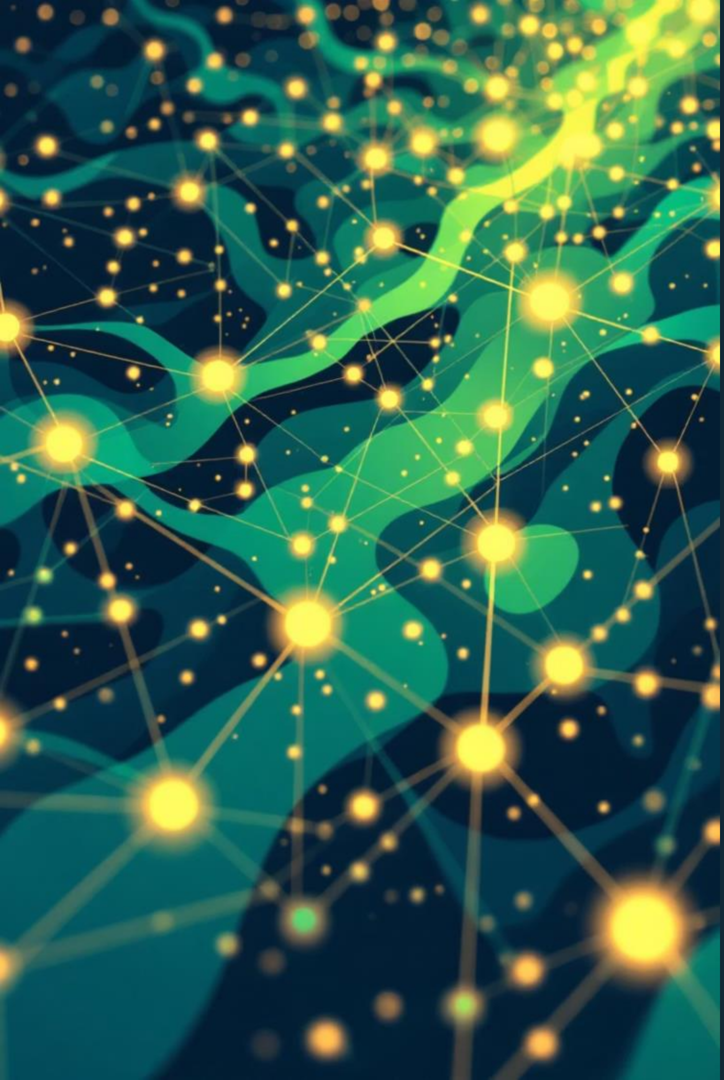
A subsequent scaling (stretching or along orthogonal axes.

3

Rotation

A final rotation to align with the output space. space.

Imagine a unit circle in a 2D plane. When a matrix transforms this circle, it becomes an ellipse. The semi-axes of this ellipse correspond to the singular values, to the singular values, defining the extent of stretching or shrinking along the principal directions. This visual analogy makes the abstract concept of matrix abstract concept of matrix transformation much more intuitive.



Why SVD Matters: Real-World Impact

The ability of SVD to decompose matrices into their fundamental building an indispensable tool across numerous domains:

- **Data Science & Machine Learning:** Essential for dimensionality reduction, feature feature extraction, and understanding complex data structures.
- **Signal Processing:** Used in noise reduction, data compression, and underlying signals.
- **Statistics:** A core component of multivariate analysis and principal component analysis.

SVD excels at reducing noise, compressing vast amounts of data without and uncovering latent patterns that are not immediately obvious. It provides rank approximations of matrices, allowing us to simplify complex data while most important characteristics.

Singular Value Decomposition (SVD)

Concepts • Computation • Worked Example

1. What is the SVD?

For any matrix A ∈ ℝ^{m×n}:

$$A = U \Sigma V^T$$

- U is $m\times m$ orthonormal ($U^TU = I$) → left singular vectors
- V is $n\times n$ orthonormal ($V^TV = I$) → right singular vectors
- Σ is $m\times n$ “diagonal-shaped” with $\sigma_1 \ge \sigma_2 \ge \dots \ge 0$
- Singular values σ_i measure how much A stretches along special directions

Dimensions (shapes)

A : m × n

U : m × m

Σ : m × n

V : n × n

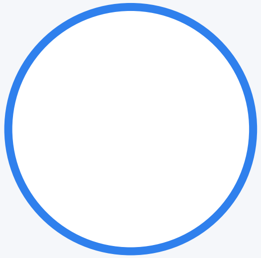
Always exists (real case → U,V orthogonal).

2. Geometric intuition: rotate → stretch → rotate

Think of A as 3 simple moves:

1) V^T

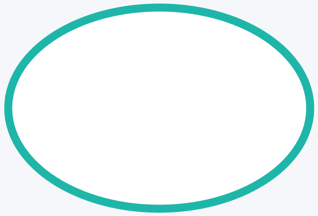
rotate / reflect input space



unit sphere

2) Σ

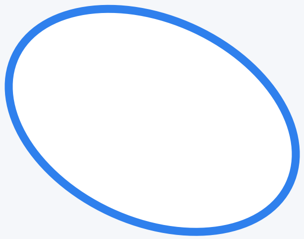
scale along coordinate axes



axes stretched by σ_i

3) U

rotate / reflect output space



ellipsoid rotated

SVD turns "messy" linear maps into orthogonal changes of basis plus simple scaling.

3. Key relationships (eigenvalues ↔ singular values)

Two symmetric matrices control the SVD:

Core identities

$A^TA = V\ (\Sigma AA^T = U\ (\Sigma\Sigma^T)\ U^T$
${}^T\Sigma)\ V^T$

Eigenvalues of AA^T and A^TA are: σ_1^2, σ_2^2, …
Left singular vectors u_i are eigenvectors of AA^T
Right singular vectors v_i are eigenvectors of A^TA

Practical consequences

- To find singular values: compute eigenvalues of AA^T (or A^TA) and take square roots.
- Nonzero singular values count = rank(A).
- Largest singular value σ_1 equals the operator 2-norm $\|A\|_2$.

Helpful formula (when $\sigma_i \neq 0$):

$v_i = (1/\sigma_i)\ A^T\ u_i$ and $u_i = (1/\sigma_i)\ A\ v_i$

4. How to compute an SVD (by hand)

A reliable classroom recipe (real matrices):

1. Compute AA^T (size m×m).
2. Find eigenvalues λ_i and orthonormal eigenvectors u_i of AA^T.
3. Set singular values $\sigma_i = \sqrt{\lambda_i}$ (keep $\sigma_i \geq 0$) and build Σ.
4. For each $\sigma_i \neq 0$, compute $v_i = (1/\sigma_i)\, A^T u_i$.
5. If $n > \mathrm{rank}(A)$, complete V with orthonormal vectors in Null(A) (these correspond to $\sigma = 0$).
6. Assemble U, Σ, V and verify $A = U\Sigma V^T$ (sign flips in matching columns are OK).

Tip: AA^T is small when m is small; A^TA is small when n is small.

5. Worked example

Example1

Find the SVD of

$$A = \begin{bmatrix} 3 & 0 \\ 4 & 0 \end{bmatrix}.$$

That is, find orthonormal U, V and diagonal Σ such that $A = U\Sigma V^T$.

Solution

Step 1) Compute $A^T A$

$$A^T = \begin{bmatrix} 3 & 4 \\ 0 & 0 \end{bmatrix} \Rightarrow A^T A = \begin{bmatrix} 3 & 4 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 25 & 0 \\ 0 & 0 \end{bmatrix}.$$

Step 2) Eigenvalues of $A^T A \rightarrow$ singular values

Eigenvalues are $\lambda_1 = 25,$ $\lambda_2 = 0.$
Singular values:

$$\sigma_1 = \sqrt{25} = 5, \sigma_2 = \sqrt{0} = 0.$$

Step 3) Right singular vectors (columns of V)

They are eigenvectors of $A^T A$.

- For $\lambda_1 = 25$: eigenvector $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.
- For $\lambda_2 = 0$: eigenvector $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

So
$$V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Step 4) Left singular vectors (columns of U)

For $\sigma_i > 0$:
$$u_i = \frac{Av_i}{\sigma_i}.$$

Compute:
$$Av_1 = A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \Rightarrow u_1 = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}.$$

5. Worked example

Since the matrix is 2×2, choose $u_2$orthonormal to u_1. One valid choice:

$$u_2 = \begin{bmatrix} -\frac{4}{5} \\ \frac{3}{5} \end{bmatrix}.$$

So

$$U = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}.$$

Step 5) Build Σ

$$\Sigma = \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix}.$$

Final SVD

$$A = U\Sigma V^T = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Example 2 (Full-rank with clean eigenvalues)

Find the SVD of

$$A = \begin{bmatrix} 3 & -3 \\ 1 & 1 \end{bmatrix}.$$

<u>Solution</u>

Step 1) Compute $A^T A$

$$A^T = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix} \Rightarrow A^T A = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 10 & -8 \\ -8 & 10 \end{bmatrix}.$$

Step 2) Eigenvalues of $A^T A$

Solve $\det(A^T A - \lambda I) = 0$:

$$\det \begin{bmatrix} 10 - \lambda & -8 \\ -8 & 10 - \lambda \end{bmatrix} = (10 - \lambda)^2 - 64 = 0.$$

5. Worked example

$$(10-\lambda)^2 = 64 \Rightarrow 10-\lambda = \pm 8.$$

So $\lambda_1 = 18, \lambda_2 = 2.$

Step 3) Singular values $\sigma_1 = \sqrt{18} = 3\sqrt{2}, \sigma_2 = \sqrt{2}.$

Step 4) Right singular vectors V(eigenvectors of $A^T A$)

For $\lambda_1 = 18$:

$$(A^T A - 18I) = \begin{bmatrix} -8 & -8 \\ -8 & -8 \end{bmatrix}.$$

Equation: $-8x - 8y = 0 \Rightarrow x = -y.$

Choose $v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, normalize: $v_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}.$

For $\lambda_2 = 2$: $(A^T A - 2I) = \begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix}.$

5. Worked example

Equation: $8x - 8y = 0 \Rightarrow x = y$.

Choose $v_2 = \begin{bmatrix}1\\1\end{bmatrix}$, normalize: $v_2 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$.

So $V = \begin{bmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{bmatrix}$.

Step 5) Left singular vectors U using $u_i = \frac{Av_i}{\sigma_i}$

Compute Av_1:

$$Av_1 = \begin{bmatrix}3 & -3\\1 & 1\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}3+3\\1-1\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}6\\0\end{bmatrix} = \begin{bmatrix}3\sqrt{2}\\0\end{bmatrix}.$$

Then $u_1 = \frac{Av_1}{\sigma_1} = \frac{\begin{bmatrix}3\sqrt{2}\\0\end{bmatrix}}{3\sqrt{2}} = \begin{bmatrix}1\\0\end{bmatrix}$.

Compute Av_2: $Av_2 = \begin{bmatrix}3 & -3\\1 & 1\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}3-3\\1+1\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}0\\2\end{bmatrix} = \begin{bmatrix}0\\\sqrt{2}\end{bmatrix}$.

5. Worked example

Then

$$u_2 = \frac{Av_2}{\sigma_2} = \frac{\begin{bmatrix} 0 \\ \sqrt{2} \end{bmatrix}}{\sqrt{2}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

So

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

Step 6) Build Σ

$$\Sigma = \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}.$$

Final SVD

$$A = U\Sigma V^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Teaching note: here the singular values are very clean $(3\sqrt{2}, \sqrt{2})$, so students clearly see that σ_1 carries most "energy" → compression idea.

Question :1

Consider the linear system $Ax \approx b$ where

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1.001 \end{bmatrix}, x_{\text{true}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

1. Compute the noise-free measurement vector $b = Ax_{\text{true}}$.
2. A noisy measurement is obtained:

$$b_{\text{noisy}} = \begin{bmatrix} 2.001 \\ 2.000 \end{bmatrix}.$$

Solve for x using the direct linear solve $x = A \backslash b_{\text{noisy}}$.

3) Compute the singular values σ_1, σ_2 of A and the condition number $\kappa = \sigma_1/\sigma_2$.
4) Explain (using SVD) why a small noise in b causes a large error in x.
5) Use **Truncated SVD (TSVD)** with threshold $\tau = 10^{-2}$(drop any $\sigma_i < \tau$) to compute a stabilized estimate x_{TSVD}. Compare it with x_{true}.

Answer

(1) Compute $b = Ax_{\text{true}}$

$$b = \begin{bmatrix} 1 & 1 \\ 1 & 1.001 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+1 \\ 1+1.001 \end{bmatrix} = \begin{bmatrix} 2 \\ 2.001 \end{bmatrix}.$$

6. SVD Application Examples

(2) Direct solution with noisy data

Given $b_{\text{noisy}} = \begin{bmatrix} 2.001 \\ 2.000 \end{bmatrix}$,

the direct solve gives: $x = A^{-1} b_{\text{noisy}} \approx \begin{bmatrix} 3.001 \\ -1.000 \end{bmatrix}$.

Comment: This is far from $x_{\text{true}} = [1 \;\; 1]^T$ even though the noise in b is only about 0.001.

(3) Singular values and condition number

Compute SVD: $A = U \Sigma V^T, \Sigma = \operatorname{diag}(\sigma_1, \sigma_2)$.

6. SVD Application Examples

Numerically: $\sigma_1 \approx 2.000500, \sigma_2 \approx 0.000499875.$

So, the condition number is $\kappa = \frac{\sigma_1}{\sigma_2} \approx \frac{2.000500}{0.000499875} \approx 4002.$

(4) SVD explanation (why noise explodes)

The pseudoinverse solution is $x = A^+ b = V\Sigma^+ U^T b,$

Where $\Sigma^+ = \mathrm{diag}\left(\frac{1}{\sigma_1}, \frac{1}{\sigma_2}\right).$

Because σ_2 is very small: $\frac{1}{\sigma_2} \approx \frac{1}{0.000499875} \approx 2000,$

any noise component in the direction corresponding to σ_2 is multiplied by about **2000×**, producing a large error in x.

Interpretation: The columns of A are nearly dependent (almost the same), so one parameter direction is poorly observable and becomes noise-amplifying when inverted.

(5) TSVD stabilized solution

With TSVD threshold $\tau = 10^{-2}$, we **drop** $\sigma_2$because $\sigma_2 < \tau$.
So we invert only σ_1. The stabilized estimate becomes:

$$x_{\text{TSVD}} \approx \begin{bmatrix} 0.99975 \\ 1.00025 \end{bmatrix} \approx \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Conclusion: TSVD removes the noise-amplifying direction (tiny singular value), giving a stable solution close to the true parameters.

6. SVD Application Examples

Question :2

A real RGB image $I$of size $m \times n \times 3$is given (a colored photograph). In this question, you will use **Singular Value Decomposition (SVD)** to perform **low-rank image compression** and study the trade-off between **image quality** and **compression**.

Required

1. Convert the image to double precision in $[0, 1]$and extract the three-color channels R, G, B.
2. For each channel, compute the economy SVD:

$$R = U_R \Sigma_R V_R^T, G = U_G \Sigma_G V_G^T, B = U_B \Sigma_B V_B^T.$$

3. For a chosen rank r, construct the rank-r approximation of each channel:

$$R_r = U_R(:,1:r)\Sigma_R(1:r, 1:r)V_R(:,1:r)^T$$

(and similarly, G_r, B_r).
Then form the reconstructed image:

$$I_r = \mathrm{cat}(3,\ R_r,\ G_r,\ B_r).$$

6. SVD Application Examples

4. For each $r \in \{10,20,40,80\}$, compute and report:

- (a) **Relative error**

$$\text{RelErr}(r) = \frac{\| I - I_r \|_2}{\| I \|_2}$$

- (b) **PSNR** (Peak Signal-to-Noise Ratio) in dB
- (c) **Approximate compression ratio** (numbers stored)

$$\text{CR}(r) = \frac{3mn}{3(mr+r+nr)} = \frac{mn}{r(m+n+1)}.$$

5. Explain briefly (2–4 lines) the **importance of SVD** in compression and why keeping only large singular values preserves most of the visual content.

6. SVD Application Examples

Answer

(1) Image preparation

We read the image, convert to double, and separate channels:

$$I \in [0,1]^{m\times n\times 3}, R = I(:,:,1),\ G = I(:,:,2),\ B = I(:,:,3).$$

(2) Compute SVD per channel

For each channel:

$$R = U_R\Sigma_R V_R^T, G = U_G\Sigma_G V_G^T, B = U_B\Sigma_B V_B^T.$$

The singular values (diagonal of Σ) are ordered from largest to smallest and represent the energy/information content in each mode.

(3) Rank-r reconstruction

Keep only the first $r$singular values and vectors:

$$R_r = U_R(:,1:r)\Sigma_R(1:r,1:r)V_R(:,1:r)^T$$

(and similarly, G_r, B_r), then:

$$I_r = \mathrm{cat}(3, R_r, G_r, B_r).$$

(4) Compute metrics for each r

Relative error

$$\mathrm{RelErr}(r) = \frac{\| I - I_r \|_2}{\| I \|_2}.$$

PSNR

PSNR measures reconstruction quality (higher is better). Typical guidance:

- ~25 dB: visible artifacts
- ~30 dB: acceptable
- ~35 dB+: very good

Compression ratio

Original storage $\approx 3mn$numbers.
Compressed storage per channel $\approx mr + r + nr$.
For RGB, total $\approx 3(mr + r + nr)$. Hence:

(5) Short explanation (importance of SVD)

SVD sorts image information into modes ranked by singular values. Large singular values capture the dominant structure (edges, lighting, main shapes). Small singular values mainly represent fine details and noise. Therefore, keeping only the top $r$singular values achieve strong compression while preserving most visual content.

6. SVD Application Examples –MATLAB Code

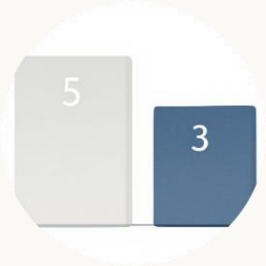
```matlab
%  Course Title : Mathematical Foundations and Software Development for AI
%  Course Code  : AI - 101
%  Academic Year: 2025/2026
%
%  Exam Example 2: SVD Compression for a Real RGB Image (Model Answer)
%
%  Written by    : A. Prof. Elsayed Mahdy
%                  Egyptian Atomic Energy Authority (EAEA)
%==========================================================================
clc; clear; close all;

%% 1) Read image and convert to double in [0,1]
imgFile = "input.jpg";          % <-- put your image name here
I = imread(imgFile);
I = im2double(I);

% (Optional) resize for faster execution in lab/class
I = imresize(I, [NaN 384]);     % keep aspect ratio, width = 384

% Get dimensions
[m,n,c] = size(I);
if c ~= 3
    error("This script expects an RGB image (3 channels).");
end

%% 2) Separate R, G, B channels
R = I(:,:,1);
G = I(:,:,2);
B = I(:,:,3);

%% 3) Compute economy SVD for each channel
[UR,SR,VR] = svd(R, 'econ');
[UG,SG,VG] = svd(G, 'econ');
[UB,SB,VB] = svd(B, 'econ');

%% 4) Ranks to test
r_list = [10 20 40 80];

% For relative error computation on whole RGB image
Ivec = I(:);
normI = norm(Ivec,2);

fprintf("\n--- SVD Image Compression Results ---\n");
fprintf("Image size: %d x %d x 3\n", m, n);
fprintf("r     RelErr         PSNR(dB)      CompressionRatio\n");
fprintf("--------------------------------------------------\n");
```

6. SVD Application Examples –MATLAB Code

```
%% 5) Loop over ranks and reconstruct
for k = 1:length(r_list)
    r = r_list(k);

    % Rank-r reconstruction per channel
    Rr = UR(:,1:r) * SR(1:r,1:r) * VR(:,1:r)';
    Gr = UG(:,1:r) * SG(1:r,1:r) * VG(:,1:r)';
    Br = UB(:,1:r) * SB(1:r,1:r) * VB(:,1:r)';

    % Rebuild the RGB image
    Ir = cat(3, Rr, Gr, Br);

    % Clip numerical rounding to [0,1]
    Ir = min(max(Ir,0),1);

    % --- Metrics ---
    relErr = norm(I(:) - Ir(:), 2) / normI;   % relative L2 error
    psnrVal = psnr(Ir, I);                    % PSNR in dB

    % Compression ratio estimate:
    % original numbers = 3*m*n
    % compressed numbers = 3*(m*r + r + n*r)
    CR = (m*n) / (r*(m+n+1));

    fprintf("%-5d %-12.6f %-11.2f %-10.2f\n", r, relErr, psnrVal, CR);

    % Display result
    figure; imshow(Ir);
    title(sprintf("Rank-%d Approx | PSNR = %.2f dB | CR=%.2f", r, psnrVal,
CR));
end

% Show original image
figure; imshow(I);
title("Original Image");
```

Intuition & Applications

Beyond the mechanics, SVD offers profound insights into data structure. The singular values and vectors reveal the most significant patterns and directions within a dataset, making SVD invaluable across various domains.



Significance of Singular Values

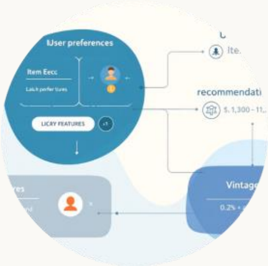
The largest singular value ($\sigma_1 = 5$) corresponds to the most significant pattern or direction in matrix A. Singular values essentially rank the "importance" of each feature or component extracted.

Orthogonal Components

SVD effectively separates data into orthogonal components, which can be interpreted as distinct features or underlying factors. This orthogonal property ensures that each component captures unique information without redundancy.

Image Compression

By truncating small singular values, SVD can achieve significant image compression. Less important components (those with smaller σ) are discarded, reducing data size while preserving visual quality.



Recommender Systems

SVD identifies latent factors that explain user preferences and item characteristics, forming the backbone of many recommender systems. It helps predict user ratings for unrated items.

Noise Reduction

In signal processing, SVD can separate noise from meaningful signals. By reconstructing the matrix using only the dominant singular values, unwanted noise components (often associated with smaller σ) are effectively filtered out.

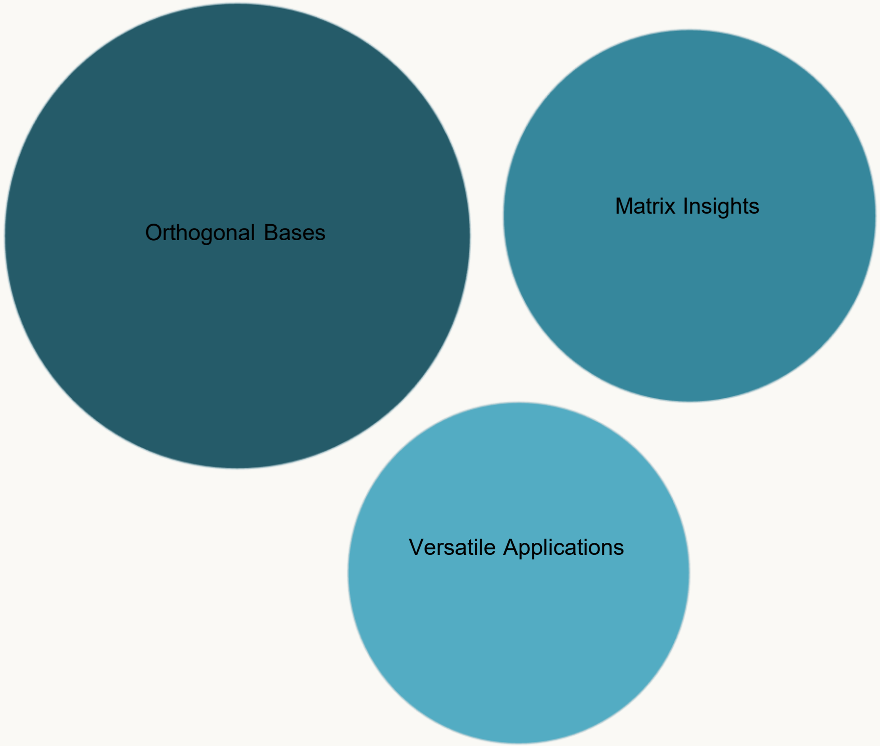
Dimensionality Reduction

SVD is a powerful tool for dimensionality reduction, including Principal Component Analysis (PCA). It projects data onto a lower-dimensional space while retaining the most variance, simplifying complex datasets for analysis.

CONCLUSION

Summary & Takeaways

Singular Value Decomposition is more than just a mathematical operation; it's a powerful framework for understanding and manipulating data. By breaking down complex matrices into simpler, interpretable components, SVD provides a unique lens through which to analyze and solve real-world problems.



الرواد الرقميون
DiGiLiANS

وزارة الاتصـــــالات
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THANK *you*